

8: Reynolds model law¹

Consider a number of similar models of the same object, differing only by their size. What can be said about the force F acting against these models, when they move through various media - liquids or gases - all keeping the same orientation relative to the direction of the velocity v . Restricting to velocities well under the velocity of sound, the density ρ of the medium is assumed to be constant - no compressions / expansions in the air occur as the model passes by.

Clearly F must depend on the *size* of the model in question. We characterise this by a *characteristic length* λ . . If, for instance, we are dealing with models of Mercedes SSK 1928, λ could be the diameter of a wheel - or a head lamp - or the length of its front bumper. In principle the choice is free, but in practice it is always understood, that one has chosen something which directly is of the same order of magnitude as the model it shall characterise. For the Mercedes, its total length is a natural choice. Or the length of the front bumper. But the diameter of a headlamp would be rather small.

The force F is also expected to depend on the *viscosity* η of the medium - a parameter which by its definition (see the text to table 8.1) measures the strength of the frictional forces which try to stop the various parts of the liquid/gas from sliding against each other. We shall assume that these parameters exempts the list, i.e.,

$$F \text{ depends on nothing but } \lambda, \rho, \eta \text{ and } v \quad (8.1)$$

Temperature	$^{\circ}\text{C}$	-32	0	18	40	54	74	230
Viscosity η	$\frac{\mu\text{N}\cdot\text{s}}{\text{m}^2}$	15.4	17.1	18.2	19.0	19.6	21.0	26.4

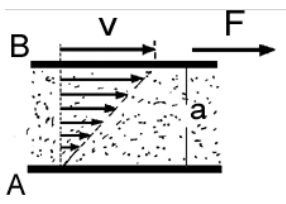


Table 8.1: The viscosity η for normal atmospheric air. In the examples considered below we use

$$\eta = 18.2 \cdot 10^{-6} \frac{\text{N}\cdot\text{s}}{\text{m}^2} \text{ as the value at } 20^{\circ}\text{C}$$

In general, the viscosity is defined as follows. Place the medium in a narrow gap of width a between two horizontal plates A and B of the same area B . The force needed to move the upper plate with a constant velocity v can then be written as $F = \eta \cdot \frac{B}{a} \cdot v$. The velocity increases linearly from 0 at the lower plate to v at the upper one.

We note that the SI-unit of η can be written in various ways

$$\frac{\text{N}}{\text{m} \cdot \text{m/s}} = \frac{\text{N} \cdot \text{s}}{\text{m}^2} = \text{Pa} \cdot \text{s} \quad \text{or} \quad \frac{\text{N}}{\text{m} \cdot \text{m/s}} = \frac{\text{N} \cdot \text{s}}{\text{m}^2} = \frac{\text{kg} \cdot \text{m/s}^2 \cdot \text{s}}{\text{m}^2} = \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

It is always possible to write the force F in terms of the dimensionless *drag coefficient* C_d defined as in (2.2) or (2.1) ,

$$F = \frac{1}{2} \rho A v^2 \cdot C_d \quad \left(\Leftrightarrow C_d = \frac{F}{\frac{1}{2} \rho A v^2} \right) \quad (8.2)$$

¹ One can read more about this law in for instance [6, p. 41-5 , 7, p. 724] and [8,9,10,11].

where A is the frontal area. Thereby the assumption of (8.1) is turned into the assumption that the dimensionless C_d is a function of (λ, ρ, η, v) . Reynolds has shown² that this function can be written as

$$C_d(\lambda, \rho, \eta, v) = C_d(R_{Rey}) \text{ where } R_{Rey} = \frac{\rho \lambda v}{\eta} \quad (8.3)$$

is now known as *Reynolds number*. That is, to determine C_d (and thereby F) it is not necessary to know λ, ρ, η and v separately, it is sufficient to know R_{Rey} .

This is Reynolds model law. Without this law, the work of making tables with numerical values of $C_d(\lambda, \rho, \eta, v)$ would be overwhelming. Now it is reduced to making just *one* table for each geometric form, namely the table of $C_d(R_{Rey})$.

We note that by writing Reynolds number as

$$R_{Rey} = \frac{v}{v_{Rey}}; \quad v_{Rey} = \frac{\eta}{\rho \lambda}, \quad (8.4)$$

it can be considered as the velocity v , measured in units v_{Rey} ; in the present notes called *Reynolds' velocity*. For normal atmospheric air at $20^\circ C$ with $\rho = 1.21 \frac{kg}{m^3}$, it can be written as

$$v_{Rey} = \frac{\eta}{\rho \cdot 1m} \cdot \frac{1m}{\lambda} = 1.50 \cdot 10^{-5} \frac{m}{s} \cdot \frac{1000mm}{\lambda} \quad (8.5)$$

The graph for spheres, with the diameter d serving as the characteristic length λ , is shown in fig. 8.1. A large number of experiments with many different sizes of spheres, moving in a variety of media, have confirmed Reynolds theory: The experimental values of C_d always fall on the same curve when plotted against R_{Rey} , namely the curve shown in fig. 8.1. There we have also plotted the points found by the merry-go-round for the ball considered in sec. 4. As one can see, they fall on the graph as expected according to Reynolds theory.

² Briefly [10 p. 15], Reynolds argument for the assertion of (8.2) goes as follows. It is expected that the analytical form of the dimensionless $C_d(\lambda, \rho, \eta, v)$ can be written as a sum of dimension less products of the form

$$c \cdot \lambda^a \cdot \rho^\beta \cdot \eta^\gamma \cdot v^\delta = c \cdot (\lambda^{a-\delta} \cdot \rho^{\beta-\delta} \cdot \eta^{\gamma-\delta} \cdot v)^\delta \quad (i)$$

where c is a dimension less constant and a, β, γ, δ are dimension less powers. Such a product can only be dimensionless if $(x, y, z) = (a - \delta, \beta - \delta, \gamma - \delta)$ satisfies the equation

$$m^x \cdot \left(\frac{kg}{m^3}\right)^y \cdot \left(\frac{kg}{m \cdot s}\right)^z = \frac{1}{m/s} \quad (ii)$$

Since kg should cancel out, we must have $z = -y$ and the equation reduces to

$$m^x \cdot \left(\frac{m \cdot s}{m^3}\right)^y = m^{-1} \cdot s^1 \Leftrightarrow m^{x+y-2} \cdot s^y = m^{-1} \cdot s^1 \quad (iii)$$

From this we get $y = 1$ ($\Rightarrow z = -y = -1$) and are left with $x = 1$. This means that the arbitrary product in (i) must be of the form

$$c \cdot \lambda^a \cdot \rho^\beta \cdot \eta^\gamma \cdot v^\delta = c \cdot (d^1 \cdot \rho^1 \cdot \eta^{-1} \cdot v)^\delta = c \cdot R_{Rey}^\delta \quad (iv)$$

This implies the result aimed at.

It should be emphasised that the points on the graph in fig. 8.1 have been determined experimentally. Solving the fundamental equations of fluid dynamics theoretically takes so much computer power that real progress in this field has only just begun [11]. Actually a tiny part of problem was solved theoretically by Stoke who already about 1850 showed that in the limit $\nu \rightarrow 0$, the force against the movement of a sphere with diameter d is given by

$$F = 3\pi\eta \cdot d \cdot v, \quad (8.6)$$

now known as *Stokes law* [7 p. 152, 8 p. 244,], Note that in this limit, the expression is independent of the mediums density ρ . This, of course, does not prevent us from writing F as in (8.2). With R_{Rey} from (8.4) one finds ,

$$F = \frac{1}{2}\rho A v^2 \cdot C_d \text{ where } C_d = 24 \cdot R_{Rey}^{-1} \quad (8.7)$$

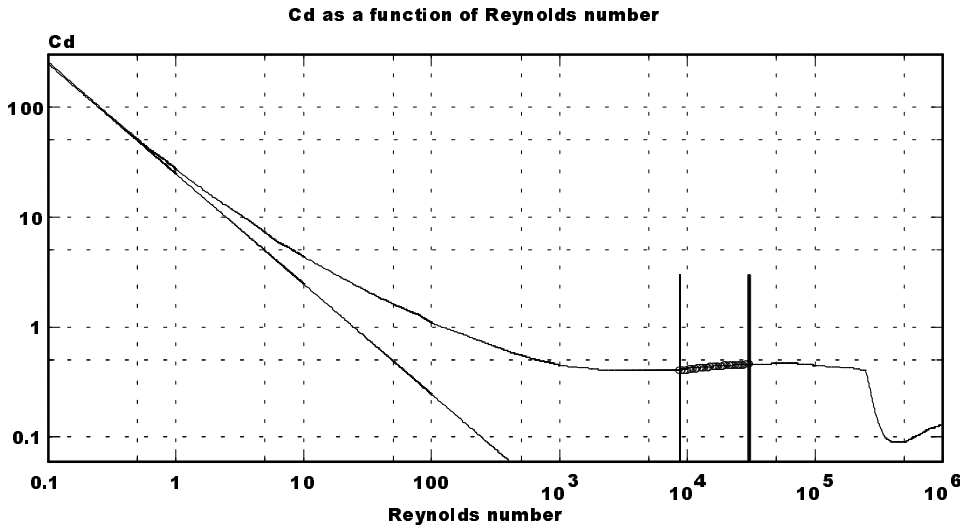


Figure 8.1³: C_d as a function of Reynolds number R_{Rey} for a Sphere. Between the two vertical lines we have also shown the experimental results $C_{d,exp}$ determined by the merry-go-round for the particular sphere considered in sec. 4 with diameter $d = 62.5 \text{ mm}$ and hence, by (8.5), $v_{Rey} = 0.240 \text{ mm/s}$. That is, we have plotted

$$C_{d,exp} = \frac{c_1}{v} + c_2 = \frac{c_1}{v_{Rey}} \frac{1}{R_{Rey}} + c_2$$

With $c_1 = -128 \frac{\text{mm}}{\text{s}}$ and $c_2 = 0.470$ as in fig. 4.2, that is $C_{d,exp} = -\frac{533}{R_{Rey}} + 0.470$

Note how the C_d -graph (in the log-log coordinatesystem) at low values of R_{Rey} follows a straight line, namely the one prescribed by Stokes law (8.7).

³ I have taken the graph in fig. 8.1 from Schlichtings book [8, p. 17] by magnifying his figure on the photo copying machine, carefully measuring the co-ordinates for a number of close-lying points by a sliding gauge, plotting these points and connecting them by straight lines. The graph in fig. 8.2, is taken in the same way from Rouse [8 p. 245]. There the final horizontal part of the curve should be at $C_d = 1.12$. I have instead inserted the value 1.17 found in the newer book [5, p. 3-17] by Hoerner.

The graph for circular disks, with the diameter d serving as the characteristic length λ , is shown in fig. 8.2. The computer program can also draw the graph of C_d for a circular cylinder, the length ℓ of which is long compared to the diameter d (the latter serving as λ). However, at present I have no experimental results.

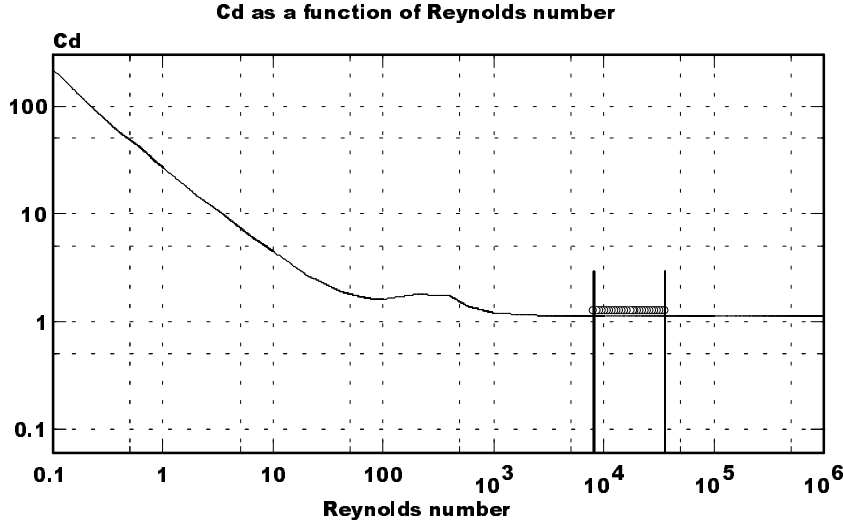


Figure 8.2: C_d as a function of Reynolds number R_{Rey} for a circular disk. Between the two vertical lines are also shown the experimental results $C_{d,exp}$ determined by the merry-go-round for the particular disk considered in sec. 4 with diameter $d = 76.0 \text{ mm}$ and hence, by (8.5), $v_{Rey} = 0.197 \text{ mm/s}$. That is, we have plotted

$$C_{d,exp} = \frac{c_1}{v} + c_2 = \frac{c_1}{v_{Rey}} \frac{1}{R_{Rey}} + c_2$$

where $c_1 = -8 \frac{\text{mm}}{\text{s}}$ and $c_2 = 1.219$ as in fig. 4.4.

Assume that we wish to use the law to find C_d for a certain body by measuring on a smaller model *moving in the same medium as the body itself*. To make sure that C_d for the model is the same as that of the real body, we must make sure that Reynolds number is the same, i.e. that

$$\lambda_{model} \cdot v_{model} = \lambda_{body} \cdot v_{body} \quad (8.8)$$

This means, that the model should move with the velocity

$$v_{model} = \frac{\lambda_{body}}{\lambda_{model}} v_{body} \quad (8.9)$$

Thus, if we wish to determine C_d for a full size Mercedes SSK going at 20 m/s ($= 72 \text{ km/h}$) by a measurement on the 1:18 model considered in sec. 7, this should be moving at $18 \cdot 20 \text{ m/s} = 360 \text{ m/s}$ ($= 1296 \text{ km/h}$). Not only is this way beyond what can be achieved on the merry-go-round, it is also well above the velocity of sound beyond which Reynolds theory no longer works. So, the good result found in sec. 7 cannot be justified simply by a reference to Reynolds model law. On the more general level, this example has tells about a limit to what can be achieved by wind tunnel measurement on small models of the real thing.

How large is Reynolds number for the full scale Mercedes SSK 1928, going at a speed of, say, $72 \text{ km/h} = 20 \text{ m/s}$?

For *the model* we get $v_{Rey} = 1.47 \cdot 10^{-4} \frac{\text{m}}{\text{s}}$ if we use length of the bumper (102 mm) as the characteristic length λ to replace d in (8.5). Since the model is built in 1:18, the value for the full scale car is

$$v_{Rey} = \frac{1.47 \cdot 10^{-4} \frac{\text{m}}{\text{s}}}{18} = 8.17 \cdot 10^{-6} \frac{\text{m}}{\text{s}} \quad (8.10)$$

Hence, at $v = 72 \text{ km/h} = 20 \text{ m/s}$,

$$R_{Rey} = \frac{v}{v_{Rey}} = 2.45 \cdot 10^6 \quad (8.11)$$

As fig. 8.3 clearly illustrates, the measurements on the merry-go-round have been carried out at much lower values of R_{Rey}

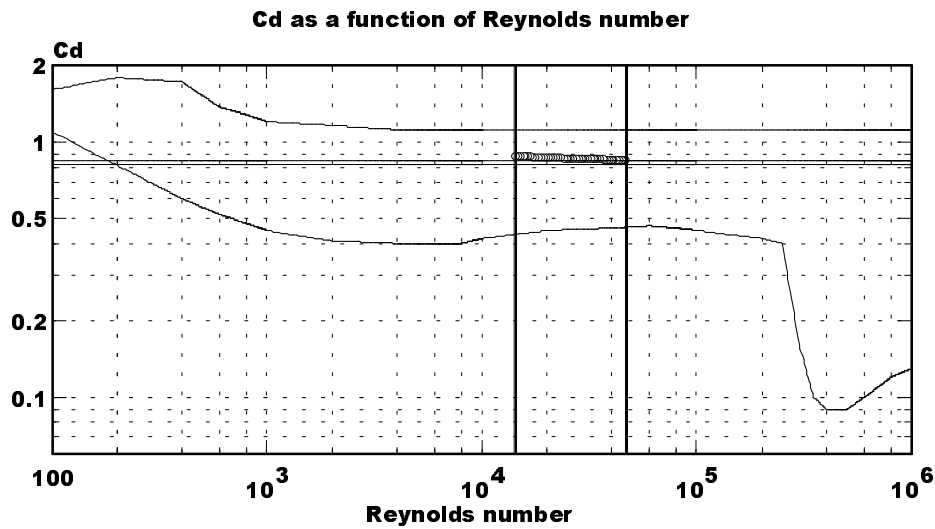


Figure 8.3: Mercedes SSK 1928. The graph between the two vertical lines shows C_d , measured on the merry-go-round, as a function of Reynolds number with the length of the front bumper as the characteristic λ . When leaving the interval to the right, this graph approaches the horizontal double line lying at the value $C_d = 0.83$, the one which in figure 7.1 is called the cars drag coefficient - without reference to the values of Reynolds number.

The two other graphs show C_d for a circular disk (upper one) and for a sphere. They are drawn just to give a crude idea of the whereabouts.